

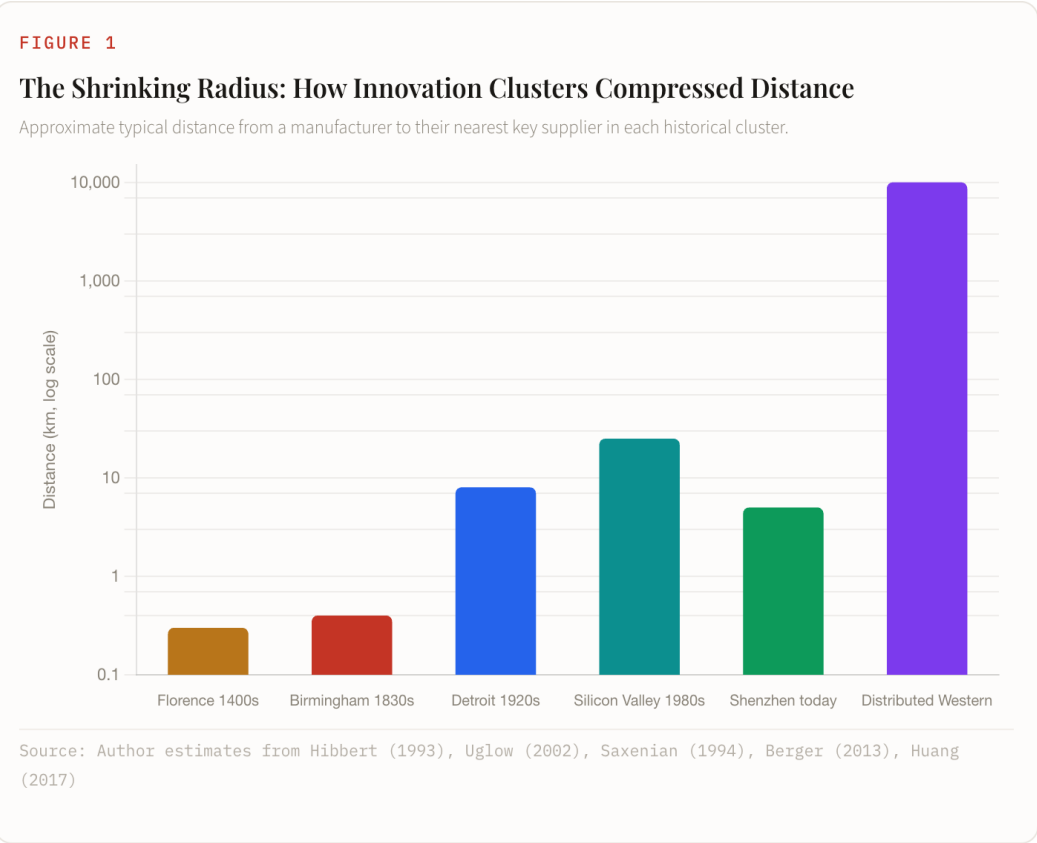
Innovation isn't genius. It's geography.

Florence in the 1400s had no monopoly on clever Italians. Venice, Milan, and Naples were full of them. What Florence had was a monopoly on closeness — silk weavers two streets from dyers, goldsmiths sharing walls with sculptors.

Gillott's weapon was a fifteen-minute walk

In 1830s Birmingham, Joseph Gillott could reach his steel roller, die cutter, stamper, and finisher on foot in minutes. He iterated pen nib designs dozens of times a year. His London rivals managed a handful. He destroyed the quill industry.

The Shrinking Radius: How Innovation Clusters Compressed Distance



From Renaissance Florence to modern Shenzhen, the winning clusters all compressed the distance between idea and execution to almost nothing.

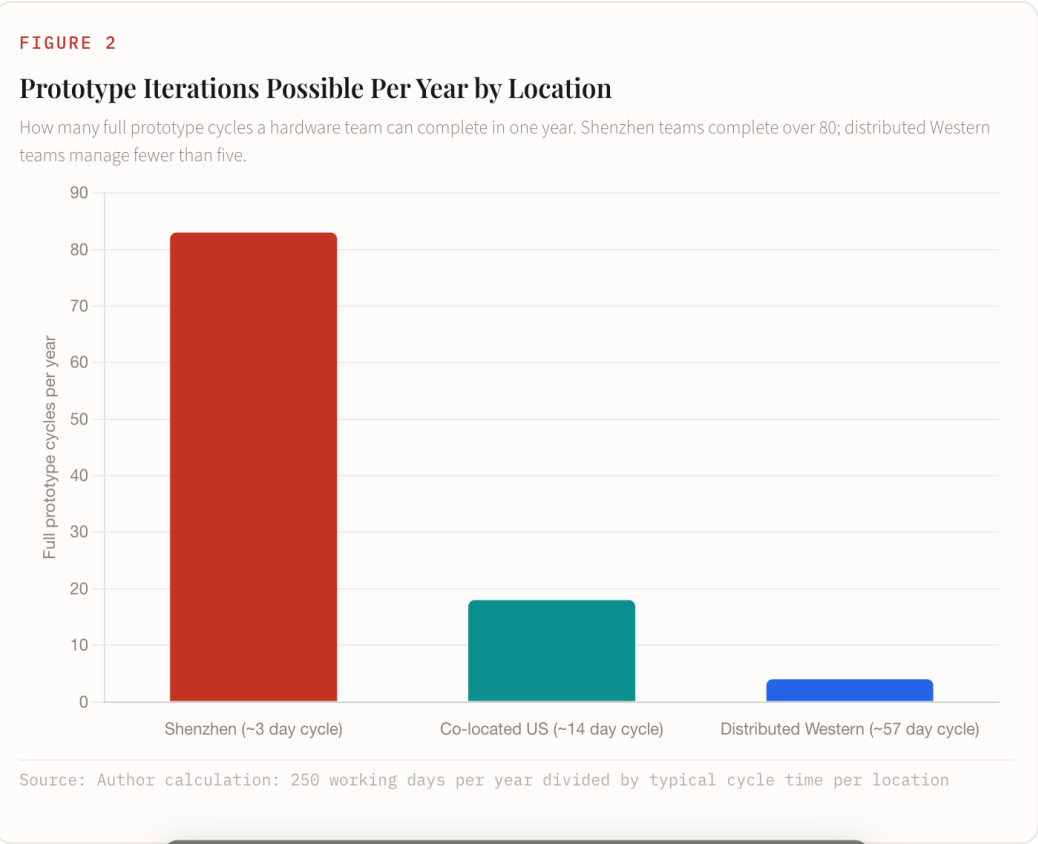
Source: Author estimates from Hibbert (1993), Uglow (2002), Saxenian (1994), Berger (2013), Huang (2017)



Proximity fakes vertical integration for free

Gillott never owned the rolling mill or the die-cutting shop. Cluster density gave him the functional equivalent of vertical integration — immediate access to every capability — without the capital burden of owning any of it.

Prototype Iterations Possible Per Year by Location

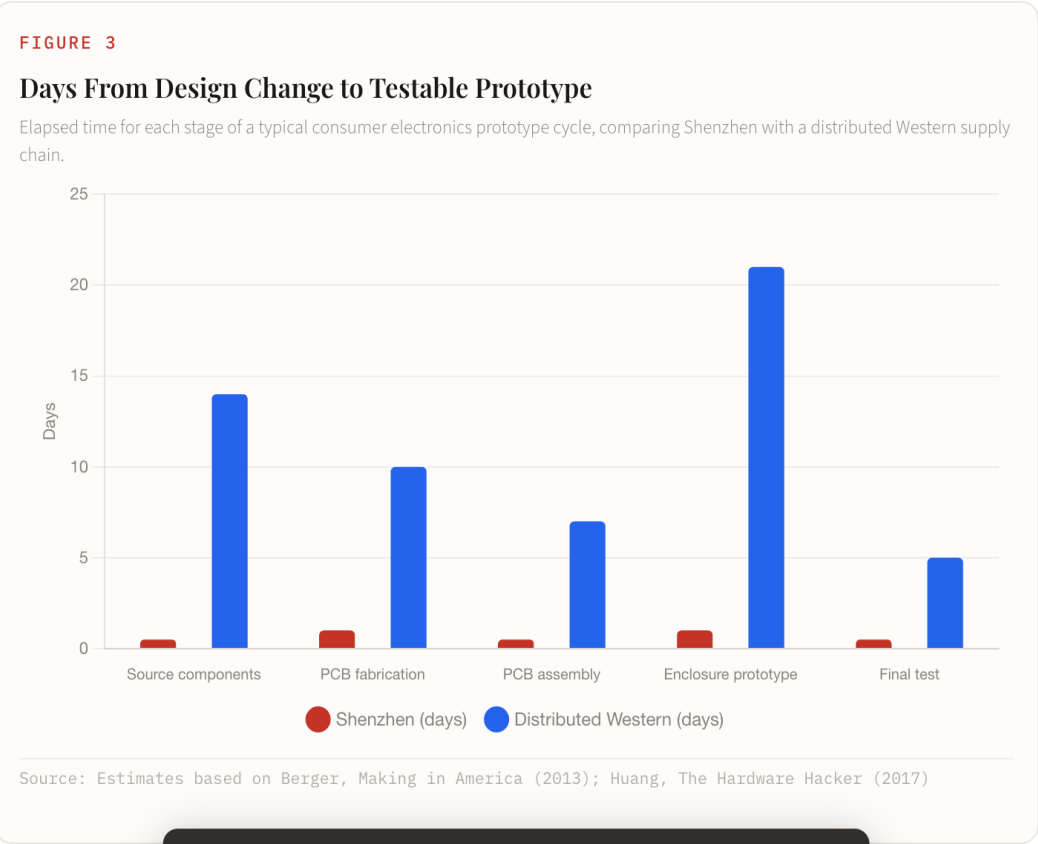


A hardware team in Shenzhen completes over 80 prototype cycles a year. A distributed Western team manages fewer than 20. That gap compounds like interest.

Source: Author calculation: 250 working days per year divided by typical cycle time per location



Days From Design Change to Testable Prototype

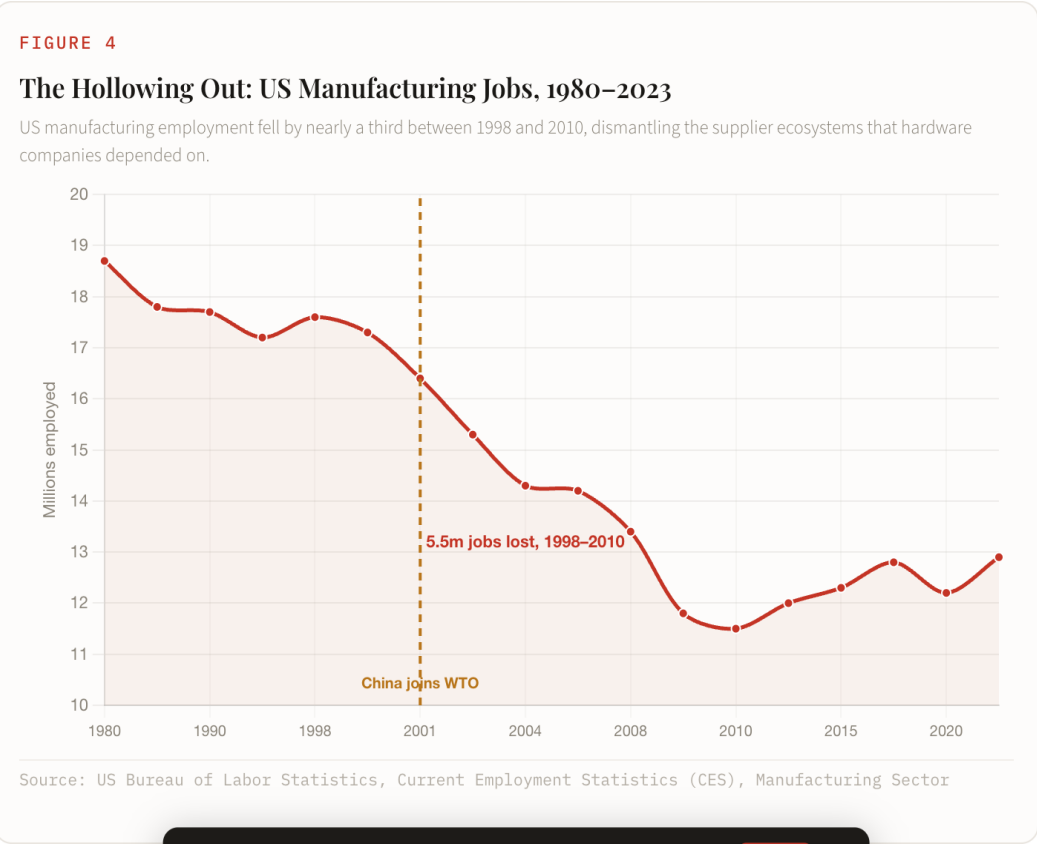


Every stage of prototyping takes longer when your supply chain is scattered across continents instead of across the street.

Source: Estimates based on Berger, Making in America (2013); Huang, The Hardware Hacker (2017)



The Hollowing Out: US Manufacturing Jobs, 1980–2023



The US lost nearly a third of its manufacturing jobs between 1998 and 2010 — and with them, the supplier ecosystems that made rapid iteration possible.

Same law, six centuries running

Florence's bottega workshops, Birmingham's pen quarter, Detroit's tool-and-die shops, Shenzhen's factory blocks: compress the distance between idea and physical test, and the speed of iteration becomes the speed of innovation. No exceptions yet.

Read the full article

The Fifteen-Minute Factory: Why Proximity Still Wins

Every innovation cluster in history succeeded because proximity collapsed the gap between idea and execution. Hardware needs that again.

<https://www.historyfuturenow.com/articles/the-fifteen-minute-factory-why-proximity-still-wins>